

Smart Biofeedback Earbud Achieved by SiP with 3D Composite Polymer Package

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Abstract— The world's first smart in-ear bud packaged in polymer is presented in this paper. This in-ear bud is capable of detecting ECG (electrocardiography) signal, heart rate, as well as body temperature, from human ear canal. The bud is designed to be connected to a typical wireless ear-set, which transmits signal through Bluetooth. It is assembled with advanced packaging technologies, including SiP (system in package) module for system board miniaturization, flexible substrate for conforming to the 3D structure, and molding with composite polymer material by multiple injections for bio contact. The composite material is liquid in raw material form, yet elastic and solid after vulcanization. This material is heterogeneous integrated, such that it encapsulates the functional active and passive parts, allows the bio signals to be detected. The earbud conforms to the exterior ear canal, and is friendly to sensitive human skin.

In this paper, this smart earbud is connected to a wearable Bluetooth docking box, and live demo is performed to transmit the detected biofeedback signal to an App on an android phone or PC. The user plugs this smart earbud into the left (heart side) ear canal and touching the metal electrode of BT docking by the left hand finger. Biofeedback signal of stable and repeatable PQRS wave, heart rate, and body temperature are detected and sent, and results are shown successfully. The test data of user trial is compared with another set of data from a leading consumer watch, tested by the same users. Very similar characteristics of the waveforms from the smart earbud, and those from the benchmark device, are observed. It is expected that this innovative smart earbud can be integrated into similar consumer wearable electronics, which can provide real time, accurate, and convenient way of bio sign monitoring for the consumers.

Keywords—electronics packaging, biosensors, electrocardiography, flexible electronics, polymers

I. INTRODUCTION

Consumer wearable devices have been among the fastest growing electronic segments in recent years. Watches, bands, wireless earphones etc. are just some of product examples that achieve fast growth and high margin[1]. Hearable devices, especially TWS(True Wireless Stereo) earphones, dominate the largest wearable market share. According to IDC[2], it is predicted that the TWS headphone market size will be 285M units in 2022 and 322M units in 2023 with averaged 15% of CAGR in 2020-2024. Driven by the 5G and beyond that promises massive connectivity on the one hand, and artificial intelligence that provides high value services to the users on the other[3], wearable electronic devices could enjoy continue

growth similar to what people witnessed of the smartphone in the coming years.

One key reason for such a rapid consumer adoption is the providing of health and bio related information to active and wellness-aware consumers. Contradicted to the desk top medical devices, consumers are demanding on the go, real time, at the spot, and affordable monitoring of their key bio signals. In addition to the basic audio and video capabilities designed for entertaining that are already in the wearable, the number of bio sensors per wearable device has been growing, and the bio measurement of the bio signals becomes more and more sophisticated. A predictable raising market on consumer wearable is transferring from fitness to healthcare functions[4]. From heart rate to ECG (electrocardiography) and from blood pressure to blood chemical density, the possibilities seem endless. As a result, these multi-function wearable devices remain high value and indispensable for the users. Apple Watch is among one of the benchmark wearable electronics that follow the trend and achieve her market status[5].

However, implement of the bio sensors on the consumer wearable electronics poses unique challenges to the device design and manufacturing. Firstly, often multiple and heterogeneous sensors – electrical, optical, vibrational, and thermal, are required in very limited available space. Secondly, the irregular human structure of the body is common for optimal sensor placement. In addition, user comfort and friendly to human skin is extremely important. Moreover, high signal fidelity and data robustness from high reliable sensor and integration requires cross discipline know-hows. Last but not least, the sensors and their integration should be of light weight, low cost, and high manufacturing through-put to meet the consumer market demands.

ASE Corporate R&D, leveraging the experience of SiP(system in package) platforms for the semiconductor devices[6], proposes a framework of solutions by using SiP technology on flexible substrates, together with conductive polymer as the innovative molding materials. SiP is ideal for integration of heterogeneous sensors and other front-end devices in a very close proximity[7,8].

To have better user experience in long duration wear and to allow that wearable sensors may work properly, we foresee a coming new approach of developing a flexible sensor platform with low power consumption [9]. Conductive polymer recently gains much attention, for examples, for biofeedback sensors of wearable for its friendliness to human skin. For the example in this paper, The 3D SiP structure with encapsulated with polymer helps to extend the bio-sensing

area from the tragus to the ear canal, which is a more desirable measuring area for bio signals, compared to current sensor solution designed with photoplethysmography (PPG) that senses at tragus[10], for example, blood oxygen saturation (SpO2). What ASE Inc. offers is a framework of solution which is unique in many ways: (1) it brings the packaging and integration of the sensors to the outside mechanical parts of consumer devices, (2) it is intended for the integration of heterogeneous sensors, which is a norm in bio sensing, and (3) the design is inherently high fidelity and consistent, as the front-end devices are directly in the same package of the sensors, and mitigate the effect of noisy and constant moving of human body parts.

Much of the effort is spend on the research of the composite polymer materials that meets the demands above. Care are giving not only to the mechanical properties of the composite polymer[11], but also to their electrical and thermal properties for the sake of the bio signal conducting. In the end, it is found that a number of key attributes of the composite materials are critical. Also, its manufacturing process, especially surface treatment at various interfaces, is critical to the yield of the integration.

II. THE SMART EARBUD: ITS STRUCTURE AND THE MANUFACTURING PROCESS

A. Structure & Features

The 3D structure of the smart earbud is achieved by SiP (system in package) and three-time injection molding of composite polymer in the assembly flow. It is composed of three parts. The inner part is a cylindrical sound tube where the SiP, integrating sensory SOC and signal processing frontend component as well as passive component and flexible carrier, is located and molded with hard composite polymer. The outer part is a soft conductive layer acts as a sensory electrode, which is made of conductive composite polymer whose electrical resistance is lower than typical polymers. There is another flexible substrate that connects SiP to the outer-layer of the sensory electrode. In order to maintain good wearable experience, a special “air cushion” is designed between the inner part and the outer electrode, which conforms to the shape of the ear canal for wearing comfort. At the bottom, an interconnecting PCB is used to connect the smart earbud with the BT docking box by pogo pins. Figure 1 shows such a structure.

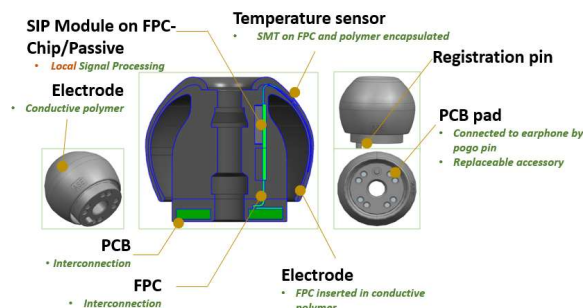


Figure 1: The cross-session view of the smart earbud structure

The functions of each of the parts are elaborated as below:

- The SiP Module on FPC: It includes one signal processing IC and other passive components that process mV-level biofeedback signal and convert to digital signal. The purpose is to minimize noise and interference for best signal fidelity and consistency.
- Temperature sensor: It is achieved by a single thermistor resistor instead of conventional infrared LED module. Taking advantage of the location of the earbud plugged in ear canal, the thermistor is positioned on the surface of the out part of the earbud, contacting ear canal for sensing body temperature. Hence the body temperature can be easily detected by the thermistor within a very short time.
- Electrode: it is made of conductive composite polymer with resistance less than $0.05 \text{ ohm} \cdot \text{meter}$. The conductivity is achieved by mixing conductive material with polymer.
- PCB: A PCB, print circuit board, is used to connect electrical signals from SiP through pogo pins with an external Bluetooth wearable box. This design is to highlight the use of earbud as a replaceable accessory of headphone, and can be swapped by the end users.

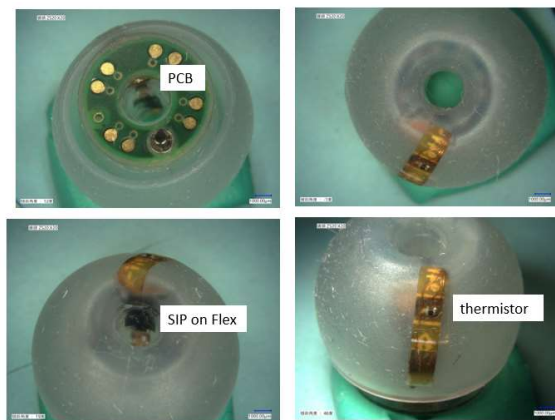


Figure 2: the top view pictures and bottom view pictures of the smart earbud structure

Figure 2 shows the top view and bottom view pictures of mockup earbud. The proto sample shown is made with transparent polymer material, in order to show the various parts. They are taken by 20X optical microscope.

B. Assembly Flow

The overall assembly process takes three injections of composite polymer to over-mold the inner part, the outer part, and the flex substrate inside of the smart earbud. The injection tooling design, choose of material for composite polymer, and injection process control are key to the yield of the assembly process.

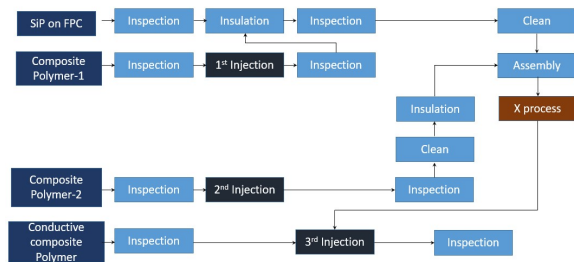


Figure 3: The assembly flow of the smart earbud

In Figure 3, each of the injection are shown with the integration process. Since the smart earbud is with a 3D sphere shape, the parts inside for assembly are of irregular shape and their placement in the assembly process also needs to be done with 3D considerations. The assembly flow starts from the assembly of “SiP on FPC”, where SiP is manufactured and surface mounted together with passive components on a flexible print circuit. Finally, The round-shape PCB is assembled by solder printing and reflow process.

The X process stands for a surface treatment process to improve the adhesive bonding among composite polymer and the other heterogeneous parts. It is one of the critical processes that may require fine tune with the choices of the polymers. It is critical for high yield, long product life cycle, and stable electrical performance.



Figure 4: pictures of the sample after the first injection, the SiP on FPC (flex print circuit substrate), and sample after the third injection as described in Figure 3

Figure 4 shows the pictures of the semi assembled parts of the smart earbud in the assembly flow as in Figure 3. The picture of 1st injection sample shows the SiP module placed in perpendicular to the wall of the cylindrical sound tube, which is hollow in middle as a sound guide for acoustic transmission from headphones. Miniaturized SiP on flex substrate allows front-end components, including both active and passive parts, to fit within the tiny space behind the wall of sound tube and be assembled vertically for accommodating the 3D ball-shape of the smart earbud. This assembly process is built upon ASE’s semiconductor SiP toolbox, and is expected to meet the challenges of the yield, cost, and scalability with the addition of the injection of composite polymer.

III. THE CONDUCTIVE COMPOSITE POLYMER MATERIAL AS ELECTRODE

Among the bio signals that are important to wearable electronics, ECG (electrical signal) and body temperature (thermo signals) are selected as the target of implementation.

ECG (electrocardiography) is a bio signal that is important to many active consumers, as well as for those consumers who need daily monitoring of cardiovascular conditions. ECG signal requires high precision of electrical measurement, and remains challenging to consumer wearable devices. Body temperature, on the other hand, becomes important with the spreading of COVID-19, and body temperature is the most widely used screening method for infection.

Material for electrode is a critical part of this research to achieve the ECG function in this smart earbud. The followings are considered as the requirements for electrode material selection:

- Non-hazardous to human body,
- Soft and comfortable for in-ear experience,
- Low viscosity for micro formation in sophisticated injection molding tools,
- Good conductive property for ECG signal level after mixing and injection, and
- Good for integrating heterogeneous sensors in one molding process.

In order to validate the conductivity of the conductive polymer, an ECG waveform test bench is set up to verify the electrical characteristic. The targeted conductive polymer is formed by injection to a solid form of strip as the DUT(device under test). This is to simulate the injection process during manufacturing. Two measures are made with the DUT, one with an ECG simulator and another from a volunteer as user. Then, the two sets of data are compared for validation. The setup is shown in Figure 5 below

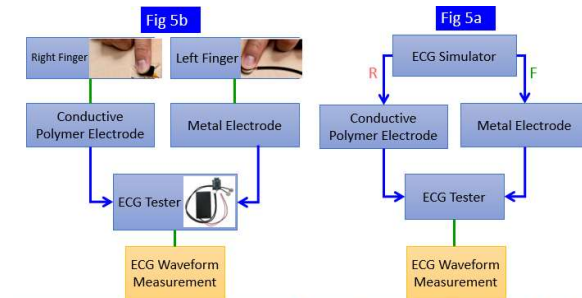


Figure 5: the test system to validate the ECG signal waveform through the conductive polymer DUT as electrode. Figure 5a is with an ECG simulator, while Figure 5b is with a volunteer

In this test system, the ECG signal passes through the DUT to an ECG tester, which processes signal and outputs ECG waveform to an application software installed on personal computer. Measurements of ECG signal from both simulator and from volunteers are made. This is to identify the effect of source impedance (could be both resistive and capacitive) to the measured waveform with the DUT as the electrode.

Many types of conductive polymer are made and tried with this test system. Several formula of composite conductive polymer, made by mixing of conductive material, proved to be promising with this test system. Figure 6 shows three typical waveforms taken from this test system.

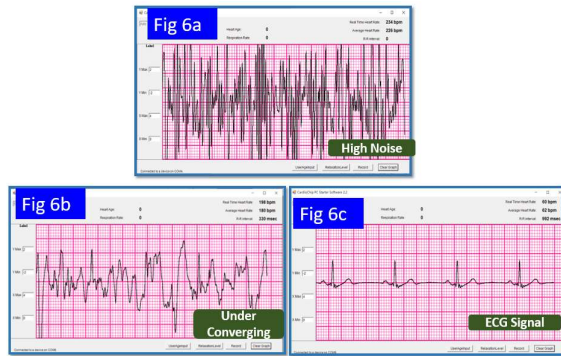


Figure 6: Three typical waveforms (ECG signal) taken from the test system in Figure 5

There are three types of typical waveforms to categorize the electrical characteristic of the conductive polymer DUT. In Figure 6a, it shows that ECG tester cannot resolve the signals through electrode. Often, it is resulted from excessive high resistance of DUT; In Figure 6b, it shows the waveform more or less repeatable in a certain test cycle time, yet there is no PQRS (ECG key signatures) waveform that can be captured. These are made with conductive composite polymers that are not good for the electrode material of the smart earbud. In Figure 6c, a good ECG waveform is shown when the conductive material of the desirable conductivity (with the sources at the signal level) of the composite polymer. The PQRS waveform can be clearly identified, and is repeatable during the whole test cycle time.

By checking the ECG waveform output from the test system, the conductivity of the polymer DUT is characterized. The composite conductive polymers that give the desirable waveforms are selected as candidates of electrode material, and are subject to further analysis and comparisons.

Three categories of material are selected as dopant, and are mixed with polymer to make the composite polymers conductive [12, 13]. They are metal, carbon tube 1 and carbon tube 2. After vulcanized to solid form, the composite polymer strips are characterized in the test system as in Figure 5. The DC and AC resistance of the strips are measured for the purpose of correlation to the ECG waveform. The result is summarized in table I.

Specimen	Size (mm)	Thickness (mm)	DC(Ω)	AC(Ω)	Fig 5 output waveform (DUT: ECG simulator)	Fig 5 output waveform (DUT: human body)
Composite polymer A	15R	0.5	0.31	0.47		
Composite polymer B	15R	0.5	190	189		
Composite polymer C	18x12	0.5	18	49		

Table 1: Composite polymers with three kinds of dopants with ECG signal output from the test system as in Figure 5

As in shown in table I, three polymer samples, one from each category, have the desirable ECG waveform as in Figure 6c. It is observed that the waveform from human body is not as smooth as of the waveform from ECG simulator. It indicates that the biofeedback signals from human body are subject to substantial noise and interference. Nevertheless, from the perspective of output ECG waveform, all three composite polymers are acceptable for further analysis.

As was mentioned earlier, in addition to the electrical properties of the composite polymers, mechanical properties relating to the manufacturing process and the encapsulated components are to be taken into considerations. Dopants not only change polymer from non-conductive to conductive, but also depending on the mixing process which changes the micro structure. Conversely, dopants impact the vulcanization condition of the injection molding, and the injection conditions also needs to be optimized, respectively to each dopant.

The best way to look into such considerations are to exam the physical structures of these composite polymers. In Figure 7, the cross-section pictures of the three composite polymer samples are shown with x200 and x2000 optical microscope.

Specimen	Dopant	Under X200 optical microscope	Under x2000 Optical microscope
Composite polymer A	Metal		
Composite polymer B	Carbon Tube1		
Composite polymer C	Carbon Tube2		

Figure 7: cross section pictures of composite polymers DUT with different dopants under x200 and x2000 optical microscope.

It is observed that the characteristic of the composite polymer A is very different from those of B and C, as the dopants are metal versus carbon tube, respectively. Also, under the x2000 optical microscope, the dopant distribution evenness is very different among composite polymer A, B, and C.

One direct consequence of the differences in the micro structure is the manufacturability of the composite polymer. Figure 8 shows the pictures of the defects found when using composite polymer A as the electrode material.

Issues found	Photo
Uneven color -it is resulted from uneven metal dopant distribution in polymer	
Delamination issue -High viscosity to fail to over-mold bottom PCB completely	

Figure 8: Defects found when using composite polymer A as the electrode material

In figure 8, the uneven color results from the metal dopant aggregation distributed over electrode. The unevenness gets worse as the injection process repeats. The root cause might be from the high viscosity of composite polymer A. The viscosity increases significantly after metal is mixed in polymer. The high viscosity issue also results in voids and

delamination among heterogeneous parts during and after formation. As a result, Composite Polymer A is excluded from selection before these failures are solved.

Another consequence of the micro structure of the composite polymer is the reliability when using it as the electrode material. Figure 9 shows the pictures of the composite polymer B under x150 optical microscope after repeated use trials. It is observed that the unevenness of the carbon tube 1 becomes so severe, that some amount of the dopant fall outs from the surface of the composite polymer B.

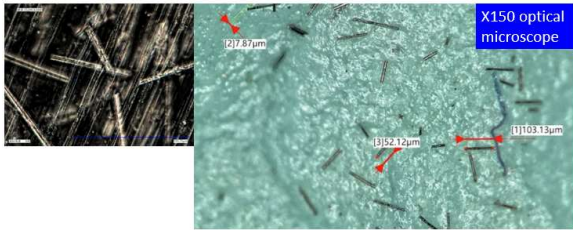


Figure 9: Pictures of the composite polymer B with carbon tube 1 after repeated user under X150 optical microscope

Such defects might cause the degradation of conductivity after long term use. Even worse, the earbud made of such polymer might leave fall-out carbon tube on user's skin, which raises serious concern as the electrode is in human's ear canal. Therefore, Composite Polymer B is dropped out from the candidate list before reliability issue and potential concern to human safety are solved.

Overall speaking, composite polymers that mix well and have good evenness in structure have the best potential as candidates for electrode materials. Figure 10 shows the SEM pictures of the Composite Polymer C with x10k, x45k, and x180k magnifications.

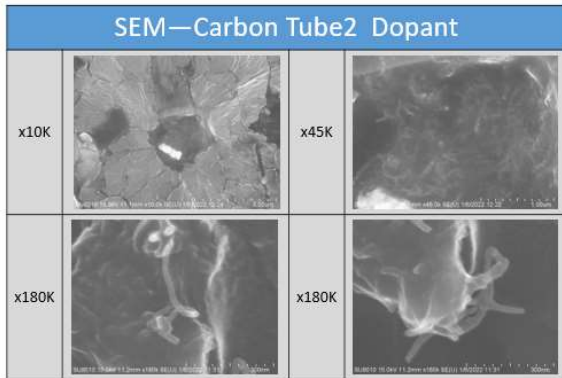


Figure 10, SEM pictures of composite polymer C with carbon tube 2 as dopant (Courtesy of Department of Material Science and Engineering, National Tsing Hua University, Hsinchu, Taiwan)

The SEM pictures show good mixtures of polymer and dopant, as well as smooth surface of the composite polymer DUT strip made of Composite Polymer C, up the 180k magnifications. There was no void found with the composite polymer, and no delamination between the polymer and the encapsulated components after manufacturing, and after repeated user trials. Therefore, the Composite Polymer C with carbon tube 2 remains in our candidate list, and its

variants are of great potential to pass the final qualification of the electrode material.

Thermal diffusivity of the composite polymer is important for thermal measurement. It is also chosen as an alternative and more accessible indication of the evenness of its micro structure and electrical conductivity. It is well known that carbon tube is a good thermal conductor as well as electrical conductor. Good and consistent thermal diffusivity is a consequence of the high uniformity of the carbon tube distribution, and hence indicates good conductivity level. Laser flash thermal diffusivity analyzer is used to obtain the value of thermal diffusivity on DUT strips of composite polymer C made in a number of manufacturing lots. Table II shows the diffusivity values of two lots of samples of composite polymer C.

Carbon Tube Doping	Test	Temperature (°C)	Diffusivity (mm ² /s)	Diffusivity Average (mm ² /s)
Sample 1	1	24.5	0.145	0.1423mm ² /s
	2	24.5	0.146	
	3	24.5	0.144	
Sample 2	1	24.5	0.142	0.145mm ² /s
	2	24.5	0.143	
	3	24.6	0.142	

Table II: Data of thermal diffusivity of composite polymer C samples from two manufacturing lots (Courtesy of Department of Material Science and Engineering, National Tsing Hua University, Hsinchu, Taiwan).

On Table II, the values of diffusivity of the two lots of samples of composite polymer C are shown to be in excellent consistence. This result gives good confidence in the yield of the electrode when composite polymer C is used as the electrode material. The process controls of the mixing of the dopant 2 are in good ranges, and hence its manufacturability in good outlook. .

IV. TESTING AND SYSTEM LEVEL TEST (SLT) RESULT

A. Smart Earbud Testing

Manufacturing testing is developed for the smart earbud, continued from the experience of the set up in Figure 5. Instead of using an ECG simulator, any arbitrary waveform generator with compatible bandwidth, which is believed to be common to many tester platforms commercially available, can be used. A helpful note is that the source impedance of the waveform generator should be such that the signal level is as reasonable as at the output of the DUT.

A unique challenge to the manufacturing testing lies in the mechanical design of the 3D test fixture to test the ball shape and elastic smart earbud. In Figure 11, such a 3D test fixture is shown.

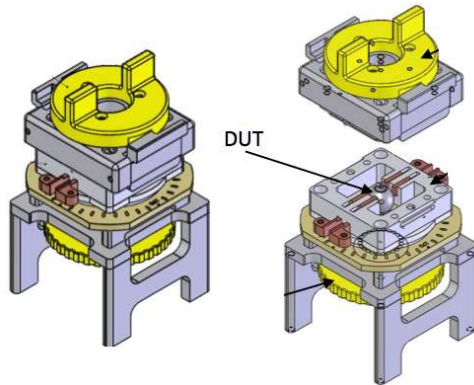


Figure 11. The innovative 3D test fixture for electrical performance testing of the smart earbud.

There are several key requirements that need to be met when designing the 3D test fixture. The angle of test pins touching the surface of the smart earbud should be perpendicular to the surface when the fixture is closed. The insertion force of the test pins should be just enough to exert the right force of contact for DC and AC test. Cares should be taken not to deform or damage the outer part of the earbud. In addition, the material used for the handler of the automatic loader should be friendly to the polymer material.

For the output of the testing, since the front-end component is inside the SiP encapsulated by the polymer, ways of testing can be considered for optimizing of the coverage vs. throughput. For example, test pins can be added to the bottom PCB of the smart earbud for straightforward DC and AC testing. Alternatively, built-in-system-test (BIST) approach can be implemented if the front-end components and SiP design allows, in order to provide better test coverage and electrical failure analysis.

B. A Complet System Level Tester

A complete tester for system level test (SLT) is developed and described as followed. It is a reference for the device OEM when the complete product using the smart earbud, a TWS (True Wireless Stereo) earphone for example, is assembled and tested. Between the smart earbud and the rest of system, an interface is built to link the biofeedback signal into device that process the waveform data. In this case, in order to best mimic the environment of a TWS ear-set, a Bluetooth docking box is developed to serve this purpose.

The Bluetooth docking box transmits biofeedback signal received from the smart earbud under test to an android phone (or any other test PC in other OS) for running analyzer software. The mechanical interface is a set of pogo pins for easy docking and replacement. This is very similar to the field usage of the smart earbud with a commercial wearable device, in which the smart earbud is used as a replaceable accessory in a plug-and-play way. The exploded view of this Bluetooth docking box is illustrated in Figure 12.

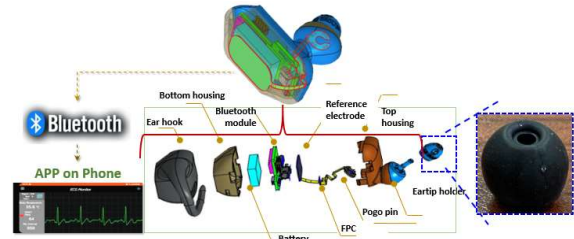


Figure 12: The exploded view of the Bluetooth docking box designed for the system level test

The functional blocks of the signal processing flow in the SLT is further illustrated in Figure 13.

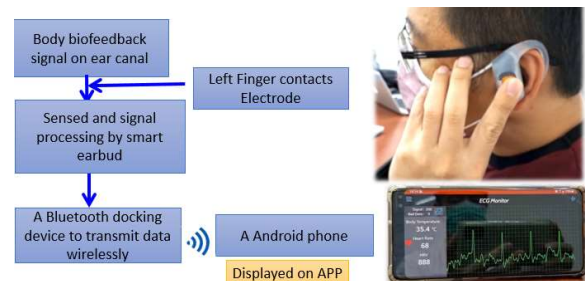


Figure 13: The functional blocks of the signal processing flow in SLT

This signal flows includes the ECG signal processing SoC ,which is built in the SiP module inside of the smart earbud, and the corresponding hardware and software functions in the docking box and the test device or PC. These hardware and software are built from off-the-shelf solutions available in many platforms. It is important to notice the correct posture of a human, when a user wearing the Bluetooth docking box is used as the ECG signal source, is critical to the test and is shown [14].

C. SLT Result

Volunteers are recruited for user trial of the SLT tester. Each user is tested by wearing the smart earbud set up as shown on Figure 13. The result from the SLT tester used on three volunteers are shown in this paper. They are male, from 40 to 43 of age, with body weight from 68Kg to 88 Kg. Immediately following the use of the SLT tester, they are tested by a benchmark consumer wearable device, and results from both tests are compared. The test results are therefore from users under similar physical conditions. The users are in resting situation, and their hear rates are between 60 to 70 per minute. For the purpose of this paper, the intention is not to apply medical clinical proven methodology on the smart earbud. Rather, the intention is to investigate the potential to use it with devices of consumer grade. Apple Watch4, which is popular among the commercially available consumer wearable devices with ECG function, is selected as the benchmark. The test results and comparison is shown in Figure 14.

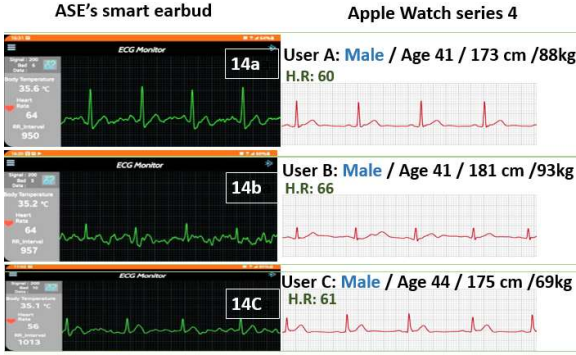


Figure 14: the ECG waveforms measured by the SLT tester with the smart earbud, and those from the benchmarking Apple Watch series 4

Figure 14 shows clear ECG waveform signatures for each of the users from both tests, and strong correlations are found between those from the smart earbud and from the benchmark consumer device. The scales on the amplitude from both tests are adjusted for ease of comparison, yet are kept the same for those from the same testers. The relative waveform peak amplitudes on 14a, 14b, and 14c are very similar to each other, and among the waveforms on each column. Closer exams on the PQRST signatures on the ECG waveforms are shown in Figure 15.

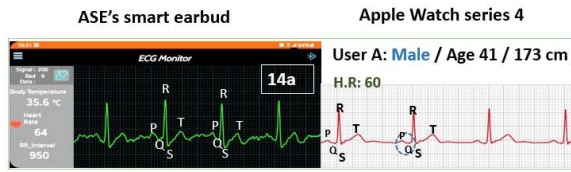


Figure 15: The ECG waveform signatures on the SLT result compared with the benchmark.

The PQRST signatures on the ECG waveform are important for the diagnostics on the cardiovascular condition of the user under test. It can be seen that all the five signature on both waveforms can be clearly identified, and their relative amplitude and time intervals are very similar to each other. This is a good indication that the smart earbud is capable of conducting reasonably accurate ECG measurement for end users, who use consumer wearable electronics as an on the go, real time, on the spot wellness monitoring device, and use the measurement as a reference for further medical cares.

D. Body Temperature Measurement Result

Infrared sensor has been widely adopted to detect the body temperature through the ear canal. Although it may reflect the result in seconds, the miniaturized infrared sensor for built-in headphone is very costly and is difficult to stably fixed in headphone, facing to ear canal without any interference from environment light. Accordingly, ASE takes another approach by embedding a thermistor on earbud surface to directly detect body temperature inside ear canal. Figure 16 illustrates the test results of four users with ear thermometer, Omron mc-523 as a benchmark for further comparison. The deviation of measurement results between smart earbud and benchmark thermometer is within 0.2 degree C, however, the response is slower. It can get 96% of measurement result within 15% test cycle time (1.5 minutes) but takes almost 85% of test cycle time (8.5 minutes) to achieve thermal equilibrium. The longer

test duration could be improved by adding more thermistors, changing the cover material on top to higher thermal diffusivity one and optimizing software algorithm...and so on. We consider this innovative approach is very promising for building in in-ear headphones for long-time continuously body temperature monitoring

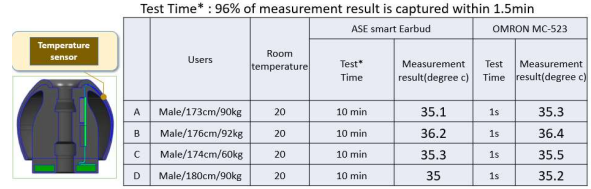


Figure 16: Four users' measurement results of body temperature tested by using ASE Smart earbud, also comparing with the ones tested by using OMRON MC-523 ear thermometer

V. CONCLUSION

A novel 3D composite polymer package is proposed in this paper. The SiP on flex substrate, manufacturing process, and encapsulating material of the smart earbud using this 3D package is developed. Manufacturing test and system level test are developed and results shown. ECG biofeedback functioning from in-ear locations [15] is demonstrated successfully with a benchmark consumer wearable electronics.

The package is unique in bringing the integration of the sensors and front-end circuitry to the outside mechanical parts of a device. This integrating approach not only reduces space and weight, but also is inherently high fidelity and high consistency for bio signal measurement.

Further studies on the structure and the composite material are being done on directions including: (1) for more heterogeneous sensors such as optical and vibrational, (2) higher reliability, and (3) other 3D form factors that fit with other human body parts. The package approach described in this paper could be applied to other consumer wearable electronics, e.g., wrist bands, patches, or glass frame, and is ideal for the integration of biofeedback signals that requires miniaturization and light weight, 3D and flexible in form factor, and friendly to human body and skin.

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